# MERCY HOUSE

A detective game that checks the evidence behind every conclusion.

## Find the truth. Show the proof.

PRESENTED BY

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# A murder. Many accounts. One investigation.

## Bhairav’s Hollow, 1900.

A physician dies while trying to expose old records. The detective knows the people involved. Some accounts are mistaken; some evidence was planted.

| 8 | 4 | 3 |
|---|---|---|
| TASKS | CASE FAMILIES | LEVELS |

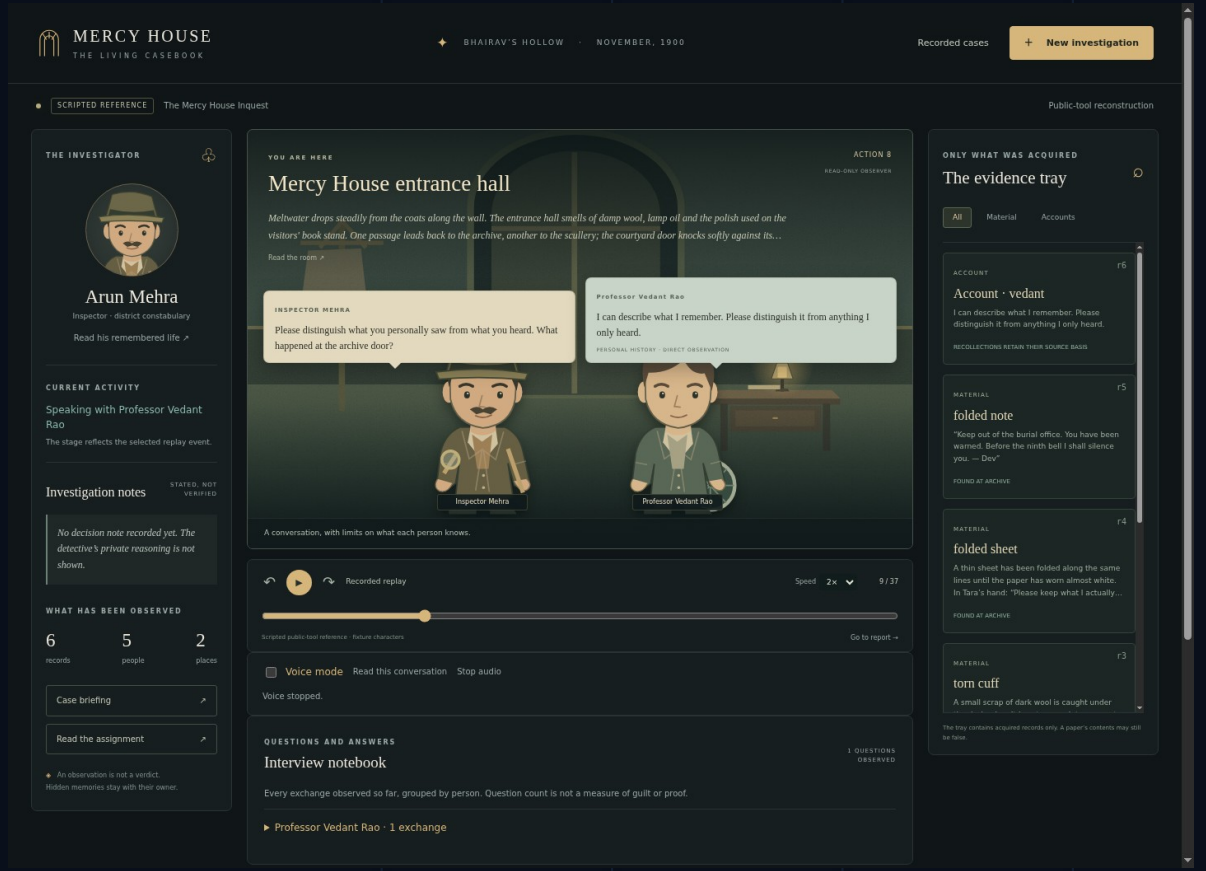



Image: a labeled recorded reference investigation. It is a demonstration, not a model-performance result.

# The detective chooses every next step.

Start at the scene with a personal history and an assignment. No solution is supplied.

### EXPLORE

Choose where to go and what to search.

### QUESTION

Ask people what they saw, heard or believe.

### CHECK

Compare documents and physical evidence.

### REPORT

Make conclusions and cite the records behind them.

The detective can be wrong, follow a false lead, change its mind or finish with unanswered questions.

Judges bring their detective model. Our submission provides the world, tools, characters and scoring.

# Actions create the clues.

The engine checks who was present, what they could do and what they could actually observe.

### THE INCIDENT

Characters move, act and handle objects.

### THE CONSEQUENCES

The engine records injuries and physical changes.

### THE WITNESSES

Each person remembers only their own experience.

Remove the assault: no fatal injury. Keep the brake intact: no coach deaths. Leave the drink unchanged: no poisoning death.

Starting incidents replay recorded or authored actions for repeatable tests. A continuously autonomous town is not implemented.

# Each person has a life beyond the question.

Family, work, relationships and private memories shape what a character is willing to say.

### WHAT THEY KNOW

Their own history and local experiences. They cannot inspect another person’s memories.

### WHAT THEY SHARE

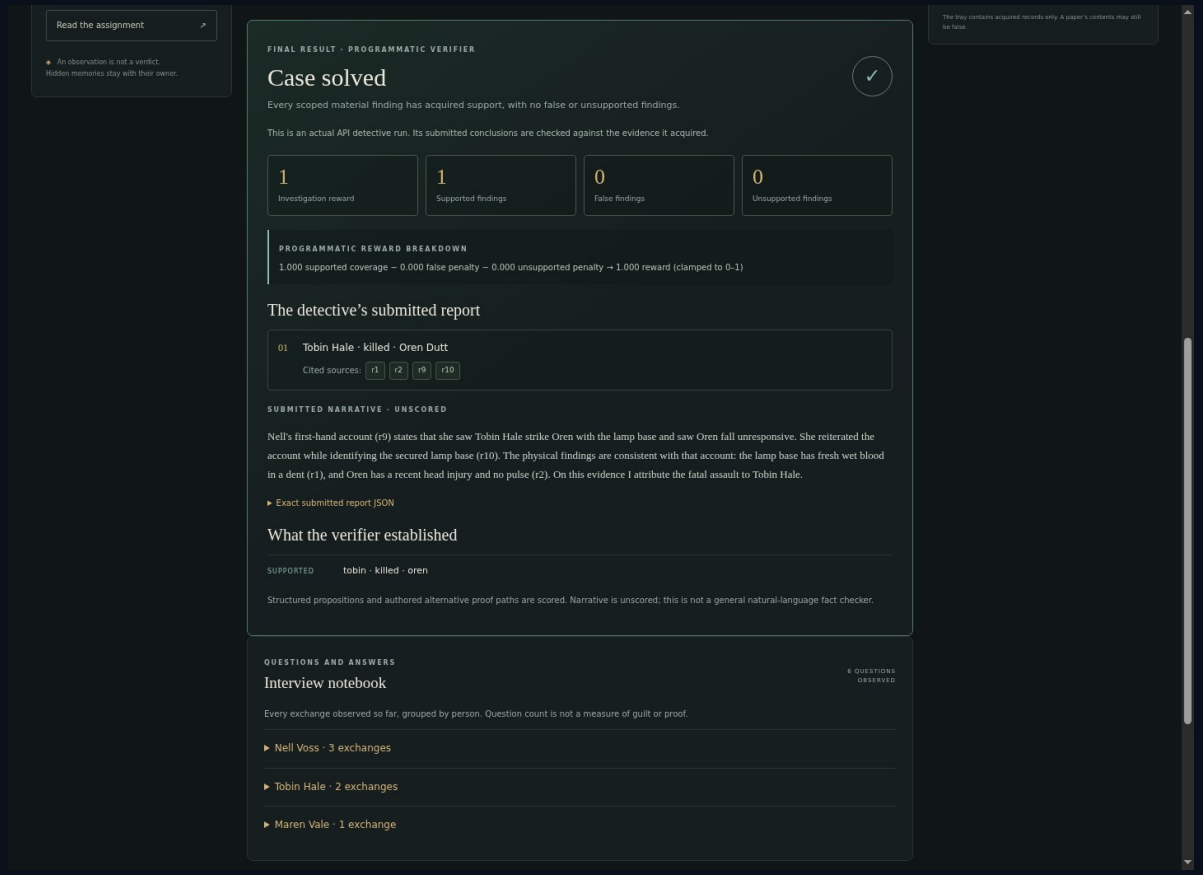
Reassurance and relevant evidence can unlock a reluctant account. A character may still refuse.

### WHAT THEY SAY

A separate model call writes the spoken reply using only the released recollections.

Witnesses do not receive the answer file. Conversation alone cannot create evidence or change what happened.

# Read the report—and why it passed or failed.




## VISIBLE RESULTS

The submitted conclusions.

Links to acquired evidence.

Supported and unsupported findings.

The score and its explanation.

Image: an actual beginner investigation that scored 1.0. The written narrative is shown, but is not automatically fact-checked.

# What we built.

## PYTHON

Runs the world, checks actions, stores evidence and scores reports.

## JAVASCRIPT + SVG

Shows places, characters, dialogue, evidence and the final report.

## DOCKER

Separates the hidden world, model access, run controller and detective’s shell.

## MODEL APIs

Power the detective and character conversations. The viewer streams their public actions.

Hosted on Google Cloud with Terraform, Cloudflare DNS and Caddy HTTPS. The public website needs no account.

# What did we train?

We trained a small action selector. We did not change the language model’s weights.

### 1 READ THE PROGRESS

Use acquired records, past actions and remaining steps.

### 2 CONSIDER CHOICES

Receive several possible searches, questions, comparisons or reports.

### 3 PICK AN ACTION

Learn which choices tend to lead to a supported conclusion.

After repeated practice, successful action sequences become more likely. This is the reinforcement-learning part.

The method is REINFORCE. Training used written rules to propose choices; the full-case test used language-model proposals.

# Three story ideas became practice problems.

We wrote new interactive exercises inspired by these stories. We did not train on complete episode scripts.

### THE MENTALIST · S1E8

Two people died. Which one was the intended target? Check the instruction and passenger records.

### THE MENTALIST · S1E7

Someone predicted a death. That does not prove murder. Check who wrote the warning.

### HIS & HERS

A familiar relative can be guilty or innocent. Use a first-hand account and physical evidence, not loyalty or gossip.

Both culprit assignments appear. Some evidence is already acquired when practice begins. The original full inquest is excluded from this new training.

# Reward proof, not a confident accusation.

## DURING PRACTICE

1 point: the requested conclusion has valid acquired evidence, with no false or unsupported claims.

0 points: otherwise.

## DURING A FULL CASE

Credit for supported required findings.

−0.20 per distinct false finding.

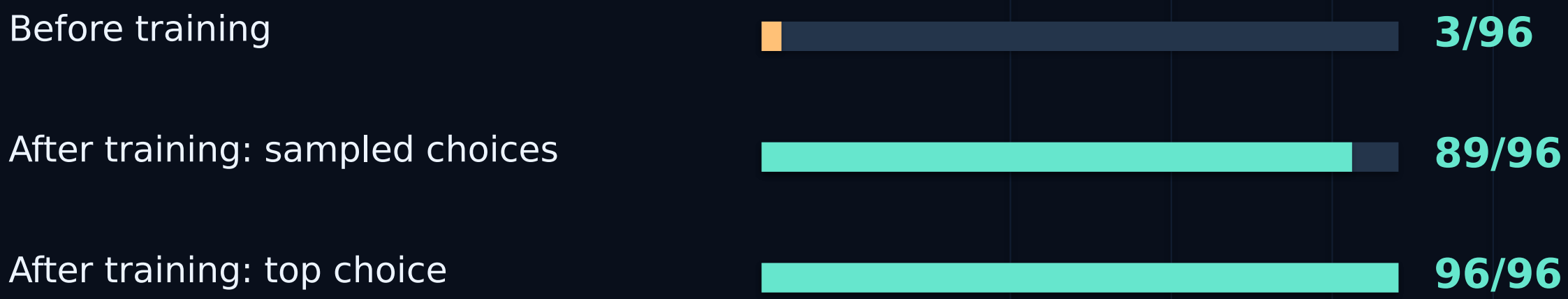
−0.10 per distinct unsupported finding.

Final score stays between 0 and 1.

A correct guess without evidence earns no credit. A Python verifier assigns the score; a language model does not grade the investigation.

# Practice improved. Full cases remain difficult.

**6,000 practice runs · 132 learned weights · 48 seconds on a CPU · no training API calls**




48 practice setups and 48 separate test setups. Each test setup was evaluated twice, using the same kinds of practice problem.

These results measure choosing among prepared actions. They do not show that a model can solve any murder case.

# The harder test exposed the gap.

We froze the new weights, then tested two tasks from the original Mercy House incident.

| | MODEL'S FIRST CHOICE | TRAINED SELECTOR |
|---|---|---|
| **Establish the fatal assault** | **0.0** | **0.0** |
| **Explain all seven material acts** | **0.0** | **0.0** |

Neither method completed the two tasks. Full-case improvement remains unproven.

Same proposal rules, fixed character responses and 40 choices per run. Four runs are a small comparison, not a broad performance claim.

# Checked, corrected and ready to inspect.

## CASE CONSISTENCY

184 task-and-seed checks passed. We fixed mixed-up roles, weak document attribution and impossible witness details.

## SOFTWARE CHECKS

124 world/runtime tests.

9 earlier regression tests.

14 viewer tests.

Desktop and mobile checks.

## TRY IT

Open the demo without signing in. Watch a recorded case, or enter your own API key to start a live investigation.

A hosted beginner test scored 1.0. Difficult live cases remain unproven; a witness can refuse to disclose crucial evidence.

# Thank you.

Scan to explore the source or watch an investigation unfold.

## SOURCE REPOSITORY



github.com/Abhishek-Punhani/mercy-house

Public source and project documentation

## LIVE DEMONSTRATION



mercyhouse.abhi07.xyz

Open access · no sign-in

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